Project Stage 2 (End-term) — Working Prototype, Data Analysis and Smart Technologies

> **Notice.** Keep the same team and the same project as registered at Project Stage 1; working alone is allowed only with the instructor's prior permission. Start the testing in Part B and the data analysis in Part C at least a week before the End-term — finding test users and fixing what they break always takes longer than expected. Agree inside the team who owns which requirement, which chart and which part of the presentation: every member presents the part they built and answers questions about it individually.

1. Purpose

Project Stage 2 is the End-term checkpoint of your trimester ICT project. At Stage 1 you designed the system. In Assignment 3 you built and protected its data layer, and in Assignment 4 you added its multimedia layer. Here you show the **working prototype** as one system, **test it against the success criteria you set at Stage 1**, **analyse the data it holds**, and apply the Week 9 topic — **smart technologies and the Internet of Things** — to your project and its sector.

Stage 2 feeds directly into the **Final Project (Week 10)**, where the completed project is submitted, presented and defended individually. Everything you report here as unfinished becomes your plan for Week 10.

Required reading

- Lecture notes for Weeks 1–9 (LMS), especially Week 9: Internet of Things, smart homes and smart grids, data analysis basics
- Nurpeisova, T. B., & Kaidash, I. N. (2017). *Information and Communication Technologies*. Almaty: Bastau — sections on smart technologies, IoT and data analysis
- Your own Project Stage 1 report and the instructor's annotations on it

> **Note.** The End-term takes place in the **Week 9** practice session, as scheduled in the syllabus. Bring the annotated Stage 1 report: several questions at the defense will check whether the problems marked there were fixed.

2. Tasks

Part A — Working prototype and requirement status (30 points)

A1. End-to-end user journey. Demonstrate the prototype completing the **main task of each user group** from Stage 1, from start to finish, with data saved in and read from your database. Include screenshots of every step in the report, in order, with a one-line caption each.

A2. Requirement status. Go through **every requirement** from your Stage 1 Part B and report its status:

ID	Requirement	Priority	Status (done / partly / not started)	Evidence (screenshot number, link or test)	Changed since Stage 1? Why?
R1	The user can register with a university e-mail address	Must	Done	Screenshot A1-2	No
R2	…	…	…	…	…

All **Must** requirements are expected to be done or partly done at this stage. For each one that is not, state the reason in one sentence.

A3. Deployment and security. State where the prototype runs — a public link, or exact instructions to run it locally — and confirm that the security measures from Assignment 3 (password handling, access rules, HTTPS) are still active. Show **one** of them working in the current version.

A4. Remaining work and contributions. Two short tables:

- **Plan to Week 10:** every unfinished requirement or known defect, who will fix it, and by when;
- **Contribution table:** one row per member stating which requirements, charts and parts of the presentation that member personally produced since Stage 1.

Part B — Testing against your success criteria (15 points)

B1. Success criteria test. Test each of your **three Stage 1 success criteria exactly as written**, with at least **three test users who are not members of your team**. Report:

Success criterion (from Stage 1)	Target	How it was tested (users, device, task)	Measured result	Met?
A student can book a room in under 60 seconds	≤ 60 s for 4 of 5 users	5 classmates, own phones, task card	4 of 5 under 60 s; mean 47 s	Yes
…	…	…	…	…

If a criterion was not met, do not change the target: report the result honestly and state what you will change before Week 10.

B2. Usability findings. List the **three most important problems** your test users met, the Nielsen heuristic each one violates, and what you changed — or will change — in response.

B3. Performance. Measure your main screen against the **performance requirement** from Stage 1 (browser developer tools, Network tab, cache disabled, "Fast 4G" throttling, at least three runs) and report the average.

Part C — Data analysis of your project (20 points)

Use the database you built in Assignment 3 — at least 100 records in the main table — to analyse what your system knows.

C1. Questions. Write **three questions** that a user or an administrator of your system would really want answered. For example: at which hours is the canteen busiest? Which equipment is booked most often? How long do document requests take on average, by document type?

C2. Method. For each question, show how you obtained the answer: the SQL query (with `GROUP BY`, `COUNT`, `AVG` or similar), or the steps in Excel or Google Sheets (pivot table, formulas). Another student must be able to repeat your analysis from what you write.

C3. Charts. Produce **at least three charts**, one per question, using the right type for the data: bar charts to compare categories, line charts for change over time, a table where exact values matter. Every chart needs a title, labelled axes with units, and a one-line note on the data source.

C4. Interpretation. In **150–200 words**, state what the data shows, **one decision** an administrator could make from it, and the **limitations** of your analysis. If your data is test data that you generated, say so openly: the charts then show that your method works, not what real users do.

C5. Data in the product. Show **at least one** of your results inside the prototype itself — for example a statistics panel on the administrator screen — with a screenshot.

Part D — Smart technologies and the Internet of Things (15 points)

D1. IoT in your sector. Choose the real sector your project belongs to — education, transport, health, food service, campus facilities, energy or another — and describe **one real example of IoT or smart technology** used in that sector, in **150–200 words**. Cover: the devices or sensors, how they connect (Wi-Fi, Bluetooth, LoRaWAN, mobile network), what data they collect, where it is processed, the benefit, and one risk. Cite at least **two sources**.

D2. A smart extension for your project. Propose **one IoT or smart feature** that would extend your system — for example an occupancy sensor at the canteen counter, an NFC or QR check-in point, a smart lock on lab equipment, or an automatic rule that notifies users when a condition is met. Provide:

- a **diagram** with the four IoT layers: device or sensor → network → cloud or processing → application, showing where it connects to your existing database;
- what data it collects, how often, and how much;
- the **security and privacy** consequence: which new personal data appears (presence, location, movement) and how you would protect it, linked to your privacy analysis from Assignment 3.

D3 (optional). Simulation. If you can, simulate the feature: for example, a short script that writes generated sensor readings into your database every few seconds, and a screen in the prototype that reacts to them. A working simulation can be shown during the defense.

Part E — Progress presentation and defense (20 points)

Present the project for **5–7 minutes** in the Week 9 session, with **every team member speaking** about the part they built, followed by **individual questions**. Use **6–10 slides**: problem and users, what works now (live demo), test results, data analysis, smart extension, plan to Week 10.

> **Note.** The demonstration must be **live**: the instructor may ask you to register a new user, run one of your analysis queries, open the developer tools, or show where a value on the screen is stored in the database. If the network or the cloud service fails, play your Assignment 4 demo video — but you will still be asked to explain and reproduce the work. Bring a laptop with the prototype, the database and your tools ready.

---

3. Deliverables checklist

☐ Report with Parts A–D, exported as a single PDF named `Group_Team_ICT_ProjectStage2.pdf`
☐ Ordered screenshots of the end-to-end user journey for each user group
☐ Requirement status table covering every Stage 1 requirement
☐ Prototype link or run instructions, and source code
☐ Success-criteria test table with the number of test users, and the three usability findings
☐ Three analysis questions, the queries or spreadsheet steps, at least three labelled charts, and one result shown inside the prototype
☐ IoT sector example with sources, and the four-layer diagram of your smart extension
☐ Plan to Week 10 and contribution table naming what each member produced
☐ Reference list (any consistent style — APA, IEEE or the style used in your textbook)
☐ Uploaded to the LMS before the deadline; one submission per team, with all members named
☐ Every member ready to demonstrate their own part live

---

4. Grading rubric (100 points)

Criterion	Excellent (90–100%)	Good (70–89%)	Satisfactory (50–69%)	Fail (0–49%)	Max
A. Working prototype and requirement status	Main task of every user group works end to end with real reads and writes; every requirement reported honestly with evidence; all Must requirements done or justified; security still active; realistic plan to Week 10	Prototype works with minor gaps; status table complete but evidence thin in places; one Must requirement missing without reason	Prototype works for one user group only, or status table incomplete or unsupported by evidence	No working prototype, or status claimed without any evidence	30
B. Testing against success criteria	All three criteria tested exactly as written with at least three non-team users; results reported honestly including failures; usability findings linked to heuristics and acted on; performance measured	Criteria tested with minor deviations from the definition, or fewer users; findings present but not acted on	Testing informal or by team members only; results asserted rather than measured	No testing	15
C. Data analysis	Three meaningful questions; repeatable method; three correct, fully labelled charts; interpretation states a decision and the limitations honestly; one result shown in the prototype	Analysis correct but questions partly trivial, or a chart poorly chosen or labelled; result not in the prototype	Charts without method, or analysis not based on the project database	No analysis, or charts invented without data	20
D. Smart technologies and IoT	Accurate sector example with sources; four-layer diagram correctly connected to the existing system; data and privacy consequences analysed concretely	Example and extension present; diagram or privacy analysis partly generic	General description of IoT not tied to the sector or the project	Missing or incorrect	15
E. Progress presentation and defense	Live demo succeeds; within time; every member presents and answers accurately, including a task asked on the spot	Good presentation; minor demo problems or uneven participation	Demo fails without recovery, or only part of the team can explain the work	Cannot demonstrate or explain the submitted work	20
				Total	**100**

The score out of 100 is converted to the **30 points** the End-term carries in the second attestation.

Formatting, citation quality and readability are assessed inside criteria A–D. Work that cannot be defended orally is graded as the defense score alone, regardless of the written quality.

In team projects the score is awarded individually where contributions differ: a member who cannot explain or reproduce the part attributed to them in the contribution table receives a reduced mark for the corresponding criteria.

---

5. Sample defense questions

1. Show me requirement R3 working right now. Which Must requirements are not finished, and why?
2. Your success criterion said 60 seconds. How many users did you test, on what device, and what did you measure?
3. Which query produced this chart? Run it now and change the grouping.
4. Your data is generated test data. What can this chart tell us, and what can it not?
5. Why is this a bar chart and not a line chart?
6. In your smart extension, which device collects the data, how does it reach the cloud, and how often?
7. What new personal data would that sensor collect, and how would you protect it?
8. What will be finished by Week 10, and which member is responsible for each item?